```
#include <stdio.h>
#include <stdlib.h>

struct node
{
   int data;
   struct node *left, *right;
};

struct node* create(int value)
{
   struct node *newnode;

   newnode = (struct node*)malloc(sizeof(struct node));
   newnode->data = value;
   newnode->left = NULL;
   newnode->right = NULL;

   return newnode;
}

struct node* insert(struct node *root, int value)
{
   if (root == NULL)
      return create(value);

   if (value < root->data)
      root->left = insert(root->left, value);
   else if (value > root->data)
      root->right = insert(root->right, value);

   return root;
}

void search(struct node *root, int key)
{
   if (root == NULL)
   {
      printf("Element not found\n");
      return;
   }

   if (root->data == key)
```

```
        printf("Element found\n");
    else if (key < root->data)
        search(root->left, key);
    else
        search(root->right, key);
}

void inorder(struct node *root)
{
    if (root != NULL)
    {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}

void preorder(struct node *root)
{
    if (root != NULL)
    {
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
    }
}

void postorder(struct node *root)
{
    if (root != NULL)
    {
        postorder(root->left);
        postorder(root->right);
        printf("%d ", root->data);
    }
}

int main()
{
    struct node *root = NULL;
    int choice, value, key;

    do
    {
```

```
printf("\n\n1. Create/Insert");
printf("\n2. Search");
printf("\n3. Inorder Traversal");
printf("\n4. Preorder Traversal");
printf("\n5. Postorder Traversal");
printf("\n6. Exit");
printf("\nEnter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        printf("Enter value: ");
        scanf("%d", &value);
        root = insert(root, value);
        break;

    case 2:
        printf("Enter element to search: ");
        scanf("%d", &key);
        search(root, key);
        break;

    case 3:
        printf("Inorder: ");
        inorder(root);
        break;

    case 4:
        printf("Preorder: ");
        preorder(root);
        break;

    case 5:
        printf("Postorder: ");
        postorder(root);
        break;

    case 6:
        printf("Exiting program\n");
        break;

    default:
        printf("Invalid choice\n");
```

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        }
    } while (choice != 6);

    return 0;
}
```